

Formulation

Properties

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Optimization: principles and algorithms



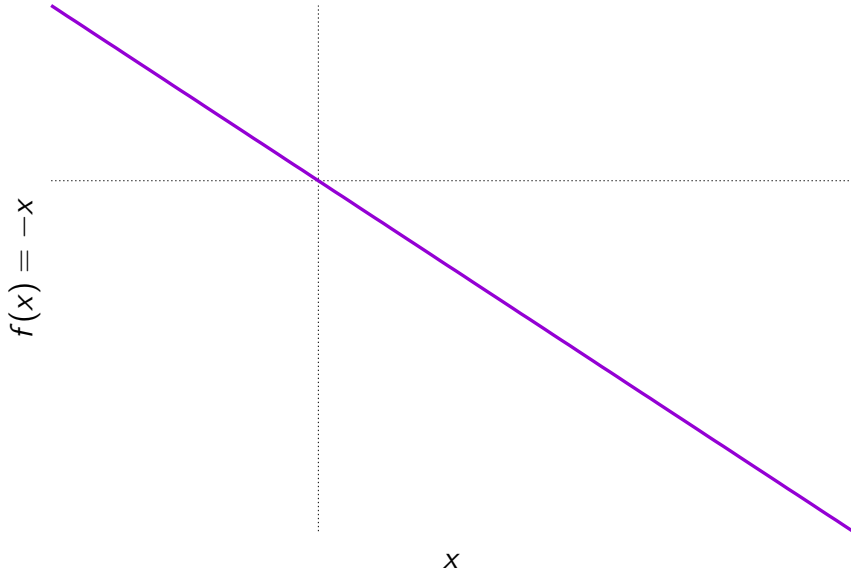
Properties

We introduce some important properties of optimization problems.

We consider from now on only minimization.

The video provides the intuition. We refer to Section 1.4 of the book for a more rigorous discussion.

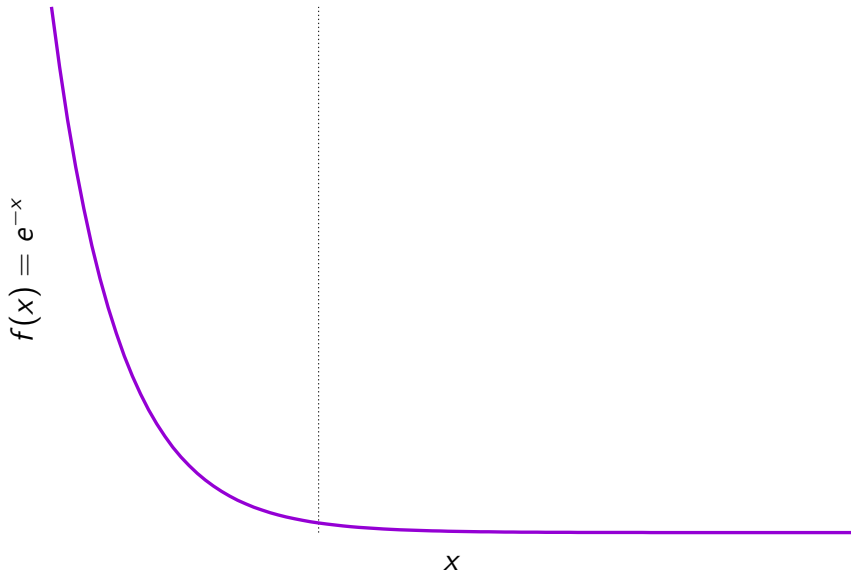
Function bounded from below



Infimum vs optimum

Explain the difference between infimum and optimum on the example. Mention Weierstrass theorem as an example of theoretical results that identify when an optimum exists.

Infimum vs optimum



Infimum vs optimum

Infimum

Largest lower bound

$$\inf_{y \in Y} f(y) \leq f(x), \forall x \in Y$$

For each $M > \inf_{y \in Y} f(y)$,

$$\exists x \in Y \text{ such that } f(x) < M.$$

Optimum

$$x^* \text{ such that } f(x^*) = \inf_{y \in Y} f(y)$$

Interior point

Problem with constraint

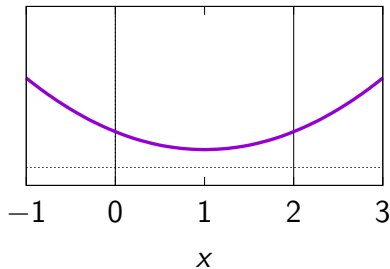
$$\min x^2 - 2x + 2$$

subject to

$$0 \leq x \leq 2$$

Problem without constraint

$$\min x^2 - 2x + 2$$



Summary

*An optimization problem is not always well defined.
It can also be simplified in some circumstances. That will be exploited.*